LECTURE NOTES FOR SURV 616

Statistical Methods II

Statistical Methods in Epidemiology

CLASS #13

OVERVIEW: In this class we will introduce some basic concepts and tools used in the analysis of Epidemiological statistics. We begin by reviewing several basic kinds of data collection methods, and the presentation of data in 2×2 tables. We review the odds ratio and relative risk, and issues about estimating them. Finally we discuss attributable risk.

References

Kahn, Harold A., and Sempos, Christopher T. (1989). *Statistical Methods in Epidemiology*. Oxford University Press.

Fleiss, Joseph L. (1981). *Statistical Methods for Rates and Proportions.* John Wiley.

Lilienfeld, David E., Stolley, Paul D. (1994). *Foundations of Epidemiology.* Oxford University Press.

1. Introduction to Data Collection and Data Presentation.

Epidemiological data is often presented in the form of the following table

Table 1. General Classification of a Population by Risk Factor and Disease Status.

	Disease Classification		
Risk Factor Classification	+(present)	-(absent)	Total at Risk
+(present)	A	B	$A+B$
-(absent)	C	D	$C+D$
Total	$A+C$	$B+D$	T

The entries in Table 1 denote the counts of individuals in the population with those attributes. We will often make use of the *capital letter notation* to denote population values. In addition, the special use of the letters A,B,C and D will be used to clearly distinguish the four cells in the tables in future examples. An example of such a table is when the Risk Factor is Smoking (Yes=+, No=-), and the disease is Lung Cancer. The group for which the risk factor is present is often called the *exposed group*, and the group for which the risk factor is absent is called the *unexposed group*.

A statistic of particular interest is called the *incidence rate* for the disease. The incidence rates for the exposed and unexposed groups are

$$(1.1) \qquad I_{Exposed} = \frac{A}{A+B}$$

$$I_{Unexposed} = \frac{C}{C+D}$$

The incidence rate for the entire population is given by

$$(1.2) \qquad I_{Pop} = \frac{A+C}{T}$$

In all cases the incidence rate is the proportion of the group (or population) in question which has the disease. The incidence rate is sometimes called the disease risk.

It is usually of primary interest to compare the incidence rates of the exposed and unexposed groups. The most natural statistics used for comparison is the *Relative Risk,* which is defined by

$$(1.3) \qquad Relative\ Risk = \frac{I_{Exposed}}{I_{Unexposed}} = \frac{\frac{A}{A+B}}{\frac{C}{C+D}} = \frac{A(C+D)}{C(A+B)}$$

The relative risk measures the incidence rate for the exposed group in *ratio* to the incidence rate for the unexposed group. In the next example we demonstrate these concepts.

Example 1. The following data is taken from an article by Abbot et al (1986) in the New England Journal of Medicine, dealing with the association between Smoking and Stroke. For the purposes of illustration we take the table below to be population values.

Data for Example 1			
	Stroke		
Smoker	Yes	No	Total
Yes	171	3264	3435
No	117	4320	4437
Total	288	7584	7872

The disease in this case is Stroke, and the Risk factor is Smoking. The incidence rates for the exposed (*Smoking* = *Yes*) and the unexposed group (*Smoking* = *No*) are given by

$$I_{Smoker} = \frac{171}{3435} \approx 0.0498$$

$$I_{Nonsmoker} = \frac{117}{4437} \approx 0.0264$$

The relative risk is given by

$$R = \frac{I_{Smoker}}{I_{Nonsmoker}} = \frac{\frac{171}{3435}}{\frac{117}{3347}} \approx 1.89$$

which says that smokers are nearly twice as likely to have a stroke as nonsmokers. Note the incidence of Stroke in the population is given by

$$I_{Population} = \frac{288}{7872} \approx 0.0366$$

which says that Strokes are a relatively rare event. Note that instead of the relative risk, we could have looked at the difference in the incidence rates and gotten

$$I_{Smoker} - I_{Nonsmoker} = \frac{171}{3435} - \frac{117}{4437} \approx 0.0234$$

In this case it looks like the difference in the incidence rates is small, but this is because the overall incidence rate is small. The relative risk gives a more useful comparison between the two incidence rates.

The relative risk is closely related to the *Odds Ratio*, also called the *Cross Product Ratio*, defined by

(1.4) $$Odds\ Ratio = \frac{AD}{BC} = \frac{\frac{A}{A+B} \Big/ \frac{B}{A+B}}{\frac{C}{C+D} \Big/ \frac{D}{C+D}}$$

The relative risk and the odds ratio are related by

(1.5) $$Relative\ Risk = \frac{\frac{A}{A+B}}{\frac{C}{C+D}} = \frac{A(C+D)}{C(A+B)} = \frac{AD\left(1+D^{-1}C\right)}{BC\left(1+B^{-1}A\right)} = Odds\ Ratio \times \frac{\left(1+D^{-1}C\right)}{\left(1+B^{-1}A\right)}$$

Therefore

(1.6) $$Relative\ Risk \approx Odds\ Ratio \quad \text{when} \quad D^{-1}C \approx B^{-1}A \approx 0$$

We would expect this approximation to work well when the incidence rate in both the exposed and unexposed groups is small.

Up to now we have been working with Table 1, which corresponds to population totals. In practice we have to work with sample data, and we next discuss the two most common types of sampling in epidemiological work.

Prospective or Cohort Studies

In prospective studies we stratify the population into the two risk factor groups, and then take a random sample from both risk groups separately. In many cases the subjects are then observed over a period of time to see whether they develop a particular disease, while in other cases the subjects are inspected for the disease. We will commonly present the data from such a study as

Table 2. Data from a Prospective (Cohort) Study.

	Disease Classification		
Risk Factor Classification	+(present)	-(absent)	Total at risk
+(present)	a	b	$a+b$
-(absent)	c	d	$c+d$

Total	$a+c$	$b+d$	t

Note the data presentation looks identical to that in Table 1, except that we have replaced the capital letters by small letters. Again we use *a,b,c* and *d* to denote the same cells as in Table 1. In a prospective study the row marginals are fixed so that

(1.7) $$a+b = \textit{fixed constant}$$

$$c+d = \textit{fixed constant}$$

and that

(1.8) $$E\left\{\frac{a}{a+b}\right\} = \frac{A}{A+B} \qquad V\left\{\frac{a}{a+b}\right\} = \frac{AB}{(a+b)(A+B)^2}$$

$$E\left\{\frac{c}{c+d}\right\} = \frac{C}{C+D} \qquad V\left\{\frac{c}{c+d}\right\} = \frac{CD}{(c+d)(C+D)^2}$$

which says that the incidence rates for the exposed and unexposed groups can be unbiasedly estimated. The relative risk is estimated by

(1.9) $$\hat{R} = \frac{\frac{a}{a+b}}{\frac{c}{c+d}} = \frac{a(c+d)}{c(a+b)}$$

If the sample sizes in the two risk groups is large, then the estimated relative risk in (1.9) is approximately unbiased for the true population risk, since the estimated relative risk is a consistent estimator of the true relative risk. Likewise the odds ratio can be estimated by

(1.10) $$\hat{O} = \frac{ad}{bc}$$

and it is also approximately unbiased for the true odds ratio, since the estimator in (1.10) is consistent.

An estimator of the variance of the log of the relative risk is given by

(1.11) $$\hat{V}\left\{\ln \hat{R}\right\} = \frac{b}{a(a+b)} + \frac{d}{c(c+d)}$$

and an estimator of the variance of the log odds ratio is given by

(1.12) $$\hat{V}\left\{\ln \hat{O}\right\} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}$$

The expressions in (1.11) and (1.12) can be used to construct approximate confidence intervals for the log relative risk, and log odds ratio.

Confidence Interval on ln(OR): $\ln(\hat{O}) \pm 1.96\sqrt{\hat{V}(\ln(\hat{O}))} = (L, U)$

Confidence Interval on OR: $(\exp(L), \exp(U))$

Retrospective or Case-Control Studies

In retrospective studies we stratify the population based on the presence/absence of the disease, and then take independent random samples from both groups. The subjects which have the disease are called the *cases* and the subjects which do not have the disease are called the *controls.* The data from a retrospective study are commonly presented in the following form

Table 3. Data from a Retrospective (Case-Control) Study.

	Disease Classification		
Risk Factor Classification	Case	Control	Total at risk
+(present)	a	b	$a+b$
-(absent)	c	d	$c+d$
Total	$a+c$	$b+d$	t

In a retrospective study, the column marginals are fixed so that

(1.13) $$a + c = \textit{fixed constant}$$

$$b + d = \textit{fixed constant}$$

Unlike a prospective study, we cannot unbiasedly estimate the incidence rates for the risk factor groups from a retrospective study since in general

(1.14) $$E\left\{\frac{a}{a+b}\right\} \neq \frac{A}{A+B}$$

$$E\left\{\frac{c}{c+d}\right\} \neq \frac{C}{C+D}$$

This is a problem which does not go away in large samples. Therefore we can not get an approximately unbiased estimator of the relative risk from a retrospective study. However, we can unbiasedly estimate the following quantities

(1.15) $$E\left\{\frac{a}{a+c}\right\} = \frac{A}{A+C} \qquad E\left\{\frac{c}{a+c}\right\} = \frac{C}{A+C}$$

$$E\left\{\frac{b}{b+d}\right\} = \frac{B}{B+D} \qquad E\left\{\frac{d}{b+d}\right\} = \frac{D}{B+D}$$

where

(1.16) $$V\left\{\frac{a}{a+c}\right\} = \frac{AC}{(a+c)(A+C)^2}$$

$$V\left\{\frac{c}{a+c}\right\} = \frac{AC}{(a+c)(A+C)^2}$$

$$V\left\{\frac{b}{b+d}\right\} = \frac{BD}{(b+d)(B+D)^2}$$

$$V\left\{\frac{d}{b+d}\right\} = \frac{BD}{(b+d)(B+D)^2}$$

Therefore if the sample sizes for both the cases and the controls are large, then the estimated odds ratio

(1.17) $$\hat{O} = \frac{ad}{bc}$$

is a consistent estimator of the true odds ratio, and the estimated odds ratio is approximately unbiased. The variance of the log-odds can be estimated by

(1.18) $$\hat{V}\left\{\ln \hat{O}\right\} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}$$

Note that the relative risk of having the risk factor is

$$\frac{\left(\frac{A}{A+C}\right)}{\left(\frac{B}{B+D}\right)} = \left(\frac{A}{B}\right)\left(\frac{B+D}{A+C}\right) = \left(\frac{AD}{BC}\right)\left(\frac{\frac{B}{D}-1}{\frac{A}{C}-1}\right) \doteq \frac{AD}{BC}$$

if the proportion of having risk factor is small among both cases and controls.

In cases where the incidence rate of the disease is rare, we know that the relative risk and the odds ratio are approximately equal. Therefore in practice, if a disease is rare researchers will estimate the relative risk by the estimated odds ratio in (1.17) and use (1.18) as an estimate of the variance of the estimated relative risk. This approximation can only be expected to work well for diseases with rare incidence rates.

Comparison of The Prospective and Retrospective Approaches

We have demonstrated above that both prospective and retrospective studies produce approximately unbiased estimates of the odds-ratio, while the prospective study has the advantage of also producing an approximately unbiased estimate of the relative risk. In the case of retrospective studies, the relative risk can be estimated by the odds ratio in the case of rare diseases. These are theoretical (statistical) differences between retrospective and prospective studies. There are at least two major *practical* differences between the two studies as well. We excerpt the following discussion from *Statistical Methods in Epidemiology*.

"Although prospective studies can estimate the relative risk with respect to becoming sick (incident cases), retrospective studies can usually estimate only the relative risk with respect to being sick (prevalent cases). The practical importance of this distinction ranges from trivial to enormous. The retrospective study that compares *existing* cases with non-cases (controls) on one or more potential risk factors cannot, in principle, relate to *all cases* occurring (incident) during the past *n* years because some of these incident cases may no longer be alive. The prospective study can, in principle, relate to all incident cases, whether early deaths or not, because risk factor status is measured at the onset of observation. For a disease such as myocardial infarction, in which a large fraction die promptly upon the very first clinical manifestation of the disease, relative risks derived from case-control studies *may* be quite different from those obtained in prospective studies. *If* early myocardial infarction deaths are unrepresentative of the myocardial infarction prevalent cases with respect to the risk factor being investigated, the difference can be great. In other words, if duration of disease is related to the risk factor, results from a retrospective study can be biased compared to a prospective study. Furthermore, a retrospective study tends to identify factors to improved survival as risk factors for being sick. On the other hand, relative risk for a disease such as senile cataract, which is not associated with increased mortality risk, is, in principle, equally estimable in prospective an case-control studies.

A second major qualification for retrospective studies to be able to provide estimates of relative risk equivalent to those from a prospective study relates to when risk factor information is collected. In prospective studies, these data are collected prior to the development of disease and thus cannot be influence or biased by disease status. (The possibility that ascertainment of disease in a prospective study may be biased by knowledge or risk factor status is a separate problem to be guarded against in study procedures.) In retrospective studies, risk factor data are collected after disease status is known and are subject to potential bias for that reason. Similar to differences arising from the use of incident or prevalent cases, the practical consequences of this potential problem vary greatly depending on the specific disease and risk factor under consideration. To illustrate, the [following] table lists several specific risk factors, in relation to myocardial infarction, showing a range of potential measurement bias from nonexistent to major."

Table from **Statistical Methods in Epidemiology**

Risk Factor	Potential measurement bias due to measuring *after* disease status is known
Age began smoking	It is possible that those with the disease may try harder than controls to recall correctly an even of many years past
Current smoking	Obvious major bias since myocardial infarction patients are almost always forbidden to smoke by attending physician. Current smoking probably does not reflect the risk factor intended for study. This is typically avoided by changing the risk factor from "current smoking" to smoking habits of "a month ago", "a year ago", "prior to you illness", etc.
Years of education completed	It is possible that those with the disease may be more (or less) given to exaggeration about a variable of social importance. However, the possibility exists of locating official records, thereby completely eliminating measurement bias.
Blood type	Bias is nonexistent for any objectively measurable genetic trait.

We have outlined a large number of reasons why retrospective studies have potential problems which prospective studies do not. The obvious answer is "Why do retrospective studies instead of prospective studies?". One very powerful answer is the retrospective studies are often more cost effective. To illustrate this we present the following tables taken from *Statistical Methods in Epidemiology.*

Sample Size Requirements for Prospective and Retrospective Studies

			Sample size needed in each group	
Disease incidence in unexposed group	Frequency of attribute in population (%)	Detectable relative risk	Prospective	Retrospective
1/1000	50	1.2	576,732	2,535
		2.0	31,443	177
		4.0	5,815	48
1/100	50	1.2	57,100	2,535
		2.0	3,100	177
		4.0	567	48
1/10	50	1.2	5,137	2,535
		2.0	266	177
		4.0	42	48

It is clear that significantly larger sample sizes can be required in prospective studies with rare diseases.

A Third (but unusual) Type of Study

We have outlined the two main data collection methods for epidemiological studies. A third type of data collection, though rarely done, is interesting to describe because it modifies the retrospective study approach to allow it to estimate the relative risk even when the disease is not rare. The idea is to take a sample of cases as before, but now define your controls to be a sample of the entire population (regardless of whether they have the disease). Now construct the following table of counts

Table 3. Data from a Modified Retrospective Study.

	Disease Classification		
Risk Factor Classification	Case (Disease +)	Control (Entire Population)	Total at risk
+ (present)	a	b	$a+b$
- (absent)	c	d	$c+d$
Total	$a+c$	$b+d$	t

In this form of the study

$$(1.19) \qquad \frac{a}{c} \quad \text{estimates} \quad \frac{A}{C}$$

$$\frac{b}{d} \quad \text{estimates} \quad \frac{A+B}{C+D}$$

Therefore the odds ratio

$$(1.20) \qquad \frac{a/c}{b/d} \quad \text{estimates} \quad \frac{\dfrac{A}{A+B}}{\dfrac{C}{C+D}}$$

which is the relative risk.

2. Attributable Risk

We have been looking at measures which help us to understand the association of exposure to a risk factor to the presence of a disease. While this is useful in terms of advice to doctors and patients, we often need to look at other measures when we want to understand general issues involving *public health.* The following motivation is taken form *Statistical Methods in Epidemiology.*

> "If you "know" that activity A multiplies the risk of lung cancer by 10 and that activity B multiplies the risk of lung cancer by 20 (i.e. $\hat{R}_A = 10$ and $\hat{R}_B = 20$), is it correct to infer (assuming very narrow confidence limits on both R_A and R_B) that activity B has a greater effect on the public health than activity A? Assuming that a county health department has resources to reduce or eliminate one of these hazards but not both, is an attack on B potentially of greater benefit than one on A? Suppose A were cigarette smoking and B were uranium mining and 40 percent of the adults in the county smoke cigarettes but only 0.4 percent mine uranium. Although those who mine uranium are at very great risk, the effect of this high risk on the community is small because few individuals are exposed to the risk factor. Clearly the effect of a risk factor on community health

is related to both relative risk and the percentage of the population exposed to the factor."

To help answer the kinds of questions that are posed by the situation above, we need to discuss the concept of *attributable risk.* We will speak of attributable risk in the *exposed group,* and also in the *total population.* We will discuss these in terms of the general population Table 1

Table 4. General Classification of a Population by Risk Factor and Disease Status.

Risk Factor Classification	Disease Classification +(present)	-(absent)	Total at Risk
+(present)	A	B	$A+B$
-(absent)	C	D	$C+D$
Total	$A+C$	$B+D$	T

with the associated statistics defined earlier

(2.1) $$I_{Exposed} = \frac{A}{A+B}$$

$$I_{Unexposed} = \frac{C}{C+D}$$

$$I_{Pop} = \frac{A+C}{T}$$

$$R = \frac{I_{Exposed}}{I_{Unexposed}} = \frac{\frac{A}{A+B}}{\frac{C}{C+D}} = \frac{A(C+D)}{C(A+B)}$$

Attributable Risk in the Exposed Group

We define the excess incidence rate in the exposed group as

(2.2) $$\textit{Excess Incidence Rate in Exposed Group} = I_{Exposed} - I_{Unexposed}$$

which gives the part of the exposed incidence rate *over and above* that which would be obtained from the unexposed group. Some researchers call this the attributable risk for the exposed group. We will define the *attributable risk for the exposed group* as

$$(2.3) \qquad A_{Exposed} = \frac{I_{Exposed} - I_{Unexposed}}{I_{Exposed}} = 1 - \frac{I_{Unexposed}}{I_{Exposed}}$$

which expresses the excess incidence rate as a fraction of the incidence rate for the group. This can be interpreted as the proportion of incidence in exposed over and above incidence in unexposed. This can also be equivalently written as

$$(2.4) \qquad A_{Exposed} = \frac{R-1}{R}$$

Attributable Risk in the Total Population

We define the excess incidence rate in the population as

$$(2.4) \qquad \textit{Excess Incidence Rate in Total Population} = I_{Pop} - I_{Unexposed}$$

which gives the part of the population incidence rate *over and above* that which would be obtained from the unexposed group. Some researchers call this the attributable risk for the total population. We will define the *attributable risk for the total population* as

$$(2.3) \qquad A_{Pop} = \frac{I_{Pop} - I_{Unexposed}}{I_{Pop}}$$

which expresses the excess incidence rate as a fraction of the incidence rate for the total population. This can be interpreted as the proportion of incidence in exposed over and above the incidence in the unexposed. This can also be equivalently written as

$$(2.4) \qquad A_{Pop} = \frac{P(R-1)}{1+P(R-1)}$$

where

(2.5) $$P = \frac{A+B}{T}$$

is the proportion of the total population exposed to the risk factor. If $p = 1$, $A_{pop} = A_{Exposed}$.

In order to illustrate these concepts we present the following example.

Example 2. We take data from Shurtleff (1970) who looked at a sample of individuals from the Framingham study, with respect to Coronary Heart Disease. We present it in the following table, treating as population values for the purposes of illustration.

Coronary Heart Disease for Example 2

Risk Factor	R	P	$A_{Exposed}$	A_{Pop}
Systolic blood pressure > 180	2.8	0.02	0.64	0.03
Enlarged heart on X-ray	2.1	0.10	0.52	0.10
Cigarette Smoking	1.9	0.72	0.47	0.39

Notice that the risk group *Systolic Blood Pressure > 180* has the largest relative risk and $A_{Exposed}$ in the table, but also has the smallest A_{Pop} because p is small. Likewise the risk group *Cigarette Smoking* has the smallest relative risk and $A_{Exposed}$, but also has the largest A_{Pop}. This would say that for the Framingham community, cigarette smoking is a bigger public health problem than systolic blood pressure.

Estimation of Attributable Risk for Prospective Studies

For prospective studies we define the estimators

(2.6) $$\hat{R} = \frac{a(c+d)}{c(a+b)}$$

$$\hat{A}_{Exposed} = \frac{\hat{R}-1}{\hat{R}}$$

$$\hat{A}_{Pop} = \frac{\hat{P}\left(\hat{R}-1\right)}{1+\hat{P}\left(\hat{R}-1\right)}$$

$$\hat{P} = \frac{a+b}{t}$$

In addition we can estimate the variance of the population attributable risk by

(2.7) $$\hat{V}\left\{\hat{A}_{Pop}\right\} = \frac{ct\left[ad\left(t-c\right)+bc^2\right]}{\left(a+c\right)^3\left(c+d\right)^3}$$

Estimation of Attributable Risk for Retrospective Studies

For retrospective studies things are a little more difficult. Since we cannot directly estimate the relative risk, we have to assume that the incident rate for the disease is small so that we can use the odds ratio as an estimate of the relative risk

(2.8) $$\hat{R} = \frac{ad}{bc}$$

we then define the attributable risk for the exposed group as

(2.9) $$\hat{A}_{Exposed} = \frac{\hat{R}-1}{\hat{R}}$$

In practice it seems that most people estimate the population attributable risk under the assumption of rare disease by

(2.10) $$\hat{A}_{Pop} = \frac{\hat{P}\left(\hat{R}-1\right)}{1+\hat{P}\left(\hat{R}-1\right)}$$

$$\hat{P} = \frac{b}{b+d}$$

This estimator of the proportion P can be motivated by the fact that

(2.11) $$P = \frac{A+B}{A+B+C+D} = \left(\frac{B}{B+D}\right)\left[\frac{1+B^{-1}A}{1+(B+D)^{-1}(A+C)}\right]$$

so that

(2.12) $$P \approx \frac{B}{B+D} \quad \text{when} \quad B^{-1}A \approx 0 \quad \text{and} \quad (B+D)^{-1}(A+C) \approx 0$$

Remember that

(2.13) $$E\left\{\frac{b}{b+d}\right\} = \frac{B}{B+D}$$

for a retrospective study. We can estimate the variance of the estimated attributable risk in the population by

(2.14) $$\hat{V}\left\{\hat{A}_{Pop}\right\} = \left[\frac{c(b+d)}{d(a+c)}\right]^2 \left[\frac{a}{c(a+c)} + \frac{b}{d(b+d)}\right]$$

We next work an example to demonstrate the estimation of these statistics

Example 3. We used the following dataset taken from *Statistical Methods in Epidemiology***.** The observations deal with the presence of coronary heart disease and its relations to systolic blood pressure.

Data for Example 3

	Coronary Heart Disease		
Systolic blood pressure (mm Hg)	Yes	No	Total
> 165	95	201	296
< 165	173	894	1067
Total	268	1095	1363

We demonstrate the analysis, under the assumption that it is a prospective study, and as a retrospective study.

Prospective Study

We get

$$\hat{R} = \frac{\frac{95}{296}}{\frac{173}{1067}} = 1.97947586 = \frac{\left(\frac{A}{A+B}\right)}{\left(\frac{C}{C+D}\right)}$$

$$\hat{A}_{Exposed} = \frac{\hat{R}-1}{\hat{R}} = 0.49481576$$

$$\hat{P} = \frac{296}{1363} = 0.21716801$$

$$\hat{A}_{Pop} = \frac{\hat{P}\left(\hat{R}-1\right)}{1+\hat{P}\left(\hat{R}-1\right)} = 0.17540111$$

$$\hat{V}\left\{\hat{A}_{Pop}\right\} = 0.00197985$$

$$se\left\{\hat{A}_{Pop}\right\} = 0.03286104$$

so we report

$$\hat{A}_{Pop} = \underset{(0.033)}{0.175}$$

Retrospective Study

We get

$$\hat{R} = \frac{(95)(894)}{(173)(201)} = 2.44141222 \quad \text{(which is really the Odds Ratio)}$$

$$\hat{A}_{Exposed} = \frac{\hat{R}-1}{\hat{R}} = 0.59040100$$

$$\hat{P} = \frac{201}{1095} = 0.18356164 = \frac{B}{B+D}$$

$$\hat{A}_{Pop} = \frac{\hat{P}\left(\hat{R}-1\right)}{1+\hat{P}\left(\hat{R}-1\right)} = 0.20922861 \quad \text{(Compare this to 0.175 above.)}$$

$$\hat{V}\left\{\hat{A}_{Pop}\right\} = 0.00140927$$

$$se\left\{\hat{A}_{Pop}\right\} = 0.03754020$$

so, we report

$$\hat{A}_{Pop} = \underset{(0.038)}{0.209}$$

The retrospective analysis yields a slightly larger attributable risk for the population, with a slightly larger standard error, but basically the results are very similar.